

Day 68 – Network Address Translation (NAT)

Overview

Day 68 introduced **Network Address Translation (NAT)** and demonstrated how private networks communicate with public networks.

In previous labs:

- Day 65 introduced RIP dynamic routing.
- Day 66 introduced DHCP automatic IP assignment.
- Day 67 introduced DNS concepts and name resolution.
- Day 68 introduced NAT and simulated Internet access.

The goal of this lab was to understand how private IP addresses are translated into public IP addresses when communicating with external networks.

Objectives

By the end of this lab, I was able to:

- Understand the purpose of NAT.
 - Differentiate between private and public IP addresses.
 - Configure a simulated ISP connection.
 - Configure NAT overload (PAT).
 - Define NAT inside and outside interfaces.
 - Verify routing and NAT functionality.
 - Troubleshoot NAT-related connectivity issues.
 - Understand how NAT relates to AWS NAT Gateways and enterprise networking.
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NAT Fundamentals

What is NAT?

NAT stands for:

Network Address Translation

NAT translates:

Private IP Address

into:

Public IP Address

when traffic leaves the local network.

Example:

192.168.10.2

becomes:

209.165.200.225

when communicating with an external network.

Why NAT Exists

Private IP addresses cannot be routed on the public Internet.

Examples:

192.168.10.2

192.168.20.2

192.168.30.2

These addresses are only valid inside private networks.

NAT allows:

Multiple Devices

to share:

One Public Address

which conserves IPv4 addresses.

NAT Overload (PAT)

The type of NAT implemented in this lab was:

PAT

or

NAT Overload

This allows:

Many Internal Devices

to share:

One Public IP Address

using different port numbers.

This is the same concept used by:

- Home routers
 - Enterprise firewalls
 - AWS NAT Gateways
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Existing Network Topology

The Day 67 topology was preserved.

LAN10 LAN20 LAN30

PC0 PC1 PC2
|||
SW0 SW1 SW2
|||
R0 ----- R1 ----- R2
|
Server0

Network Expansion

A fourth router was added to simulate an ISP connection.

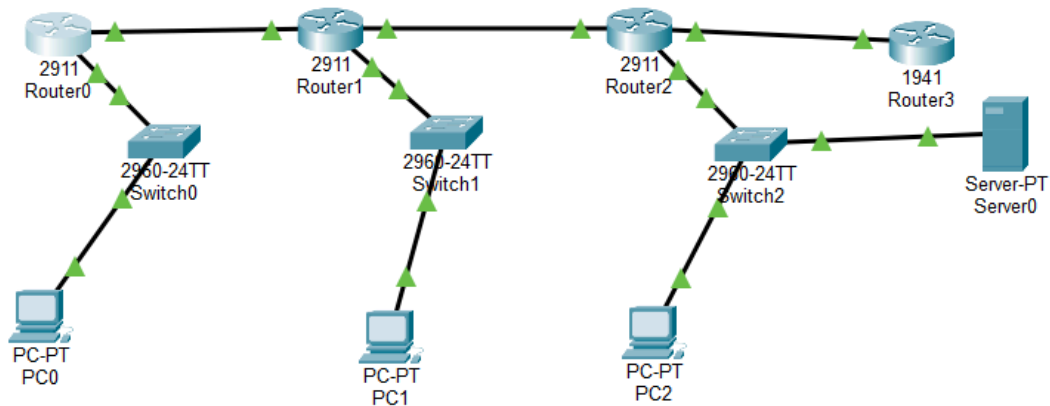
LAN10 LAN20 LAN30

PC0 PC1 PC2
|||
SW0 SW1 SW2
|||
R0 ----- R1 ----- R2 ----- Router3
|
Server0

Router3 represented:

Internet Service Provider (ISP)

Screenshot #1 – Day 68 Topology



Highlight

- Router3 added
- Existing RIP network preserved
- Server0 preserved
- New public network connection added

ISP Network Configuration

Router2

Configured a public-facing interface.

```
interface g0/2
ip address 209.165.200.225 255.255.255.252
no shutdown
```

Purpose:

Acts as the Internet-facing interface.

Router3

Configured the ISP-facing interface.

```
interface g0/0
ip address 209.165.200.226 255.255.255.252
no shutdown
```

Purpose:

Simulated external ISP network.

Screenshot #2 – Router2 Public Interface

```
Router(config)#
Router(config)#interface g0/2
Router(config-if)#ip address 209.165.200.225 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Highlight

G0/2
209.165.200.225

Screenshot #3 – Router3 ISP Interface

```
Router>enable
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 209.165.200.226 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

Highlight

G0/0
209.165.200.226

Connectivity Verification

Router2 successfully reached Router3.

Command:

ping 209.165.200.226

Result:

.!!!!

The initial failed ping occurred because Router2 first needed to learn Router3's MAC address through ARP.

Subsequent pings succeeded.

NAT Configuration

ACL Creation

```
access-list 1 permit 192.168.0.0 0.0.255.255
```

Purpose:

Identifies traffic eligible for NAT translation.

Included:

192.168.10.x

192.168.20.x

192.168.30.x

Screenshot #4 – NAT ACL Configuration

```
Router(config)#  
Router(config)#access-list 1 permit 192.168.0.0 0.0.255.255  
Router(config)#  
Router(config)#end  
Router#  
%SYS-5-CONFIG_I: Configured from console by console
```

NAT Inside Interfaces

Configured on Router2.

```
interface g0/0  
ip nat inside
```

```
interface g0/1  
ip nat inside
```

Purpose:

Identify internal network interfaces.

NAT Outside Interface

Configured on Router2.

```
interface g0/2  
ip nat outside
```

Purpose:

Identify Internet-facing interface.

Screenshot #5 – NAT Interface Configuration



```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip nat outside
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Highlight

ip nat inside

ip nat outside

NAT Overload Configuration

ip nat inside source list 1 interface g0/2 overload

Meaning:

Translate all matching private addresses
and use the public address assigned to G0/2.

The keyword:

overload

allows multiple devices to share one public IP address.

Screenshot #6 – NAT Overload Configuration

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip nat inside source list 1 interface g0/2 overload
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG I: Configured from console by console
```

Initial Problem Encountered

After configuring NAT, PC0 could not ping the ISP router.

Command:

ping 209.165.200.226

Result:

Destination host unreachable

Root Cause Analysis

Initially it appeared that NAT was failing.

However, NAT was not the problem.

The actual issue was:

Router0 did not know how to reach
209.165.200.226

The routing table only contained:

192.168.10.x
192.168.20.x
192.168.30.x

No route existed for the ISP network.

Troubleshooting

A new RIP advertisement was added on Router2.

```
router rip
network 209.165.200.0
```

Purpose:

Advertise ISP network to Router1 and Router0.

Verification

Router0 learned:

R 209.165.200.0/24 via 10.0.0.2

Meaning:

To reach the ISP network,
send traffic toward Router1.

Screenshot #7 – Updated RIP Routing Table

```

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:11, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:11, GigabitEthernet0/1
R       192.168.30.0/24 [120/2] via 10.0.0.2, 00:00:11, GigabitEthernet0/1
R       209.165.200.0/24 [120/2] via 10.0.0.2, 00:00:11, GigabitEthernet0/1

Router#

```

Highlight

R 209.165.200.0/24

Final Connectivity Test

From PC0:

ping 209.165.200.226

Result:

0% loss

Successful.

The packet path was:

PC0

↓

Router0

↓

Router1

↓

Router2

↓

Router3

This demonstrated successful private-to-public communication.

Screenshot #8 – Successful ISP Ping

```
C:\>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time<1ms TTL=252
Reply from 209.165.200.226: bytes=32 time<1ms TTL=252
Reply from 209.165.200.226: bytes=32 time<1ms TTL=252
Reply from 209.165.200.226: bytes=32 time<1ms TTL=252

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Highlight

Reply from 209.165.200.226

Key Lesson Learned

One of the most important discoveries in this lab was:

Routing happens first.

NAT happens second.

Even a correctly configured NAT router cannot translate traffic if routers do not know how to reach the destination network.

Cloud Engineering Connection

This lab directly relates to AWS networking concepts.

Networking Concept	AWS Equivalent
Router2	NAT Gateway
209.165.200.225	Elastic IP
Internal LANs	Private Subnets
Router3	Internet Gateway Simulation
NAT Overload	AWS NAT Translation

What I Learned

- NAT translates private addresses into public addresses.
 - Private IP addresses cannot be routed on the Internet.
 - NAT overload allows multiple devices to share one public IP.
 - NAT relies on proper routing.
 - Routing issues can appear similar to NAT failures.
 - RIP and NAT work together.
 - NAT Gateways in AWS use the same core concepts.
 - Troubleshooting should verify routing before assuming NAT problems.
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Reflection

Day 68 introduced Network Address Translation and demonstrated how private networks communicate with public networks. I expanded the topology by adding an ISP router and configured Router2 as a NAT boundary device. During testing, connectivity initially failed even though NAT was configured correctly. Troubleshooting revealed that the issue was routing, not NAT. After advertising the ISP network through RIP, Router0 learned the route and communication succeeded. This lab

reinforced the relationship between routing and NAT and helped connect traditional networking concepts to AWS NAT Gateways and cloud networking design.